

EDUCATION

Master’s in Computer Science, *University of Illinois Urbana Champaign, IL* August 2025 — Present
Relevant Coursework: NLP, ML & Compilers, Distributed Systems, Parallel Numerical Algorithms, Generative AI, RL

Bachelor of Science in Computer Science, *University of Wisconsin Madison, WI* September 2021 — May 2025
GPA: 3.97/4.00, Majored in Computer Science, Math, Honor Student, Dean’s List, Phi Beta Kappa Member
Relevant Coursework: Linear Algebra, Numerical Methods, Probability Theory, Operating Systems, Artificial Intelligence
Game Theory, Data Structures & Algorithms, Languages & Compilers, Computer Networks, Databases

EXPERIENCE

Applied Scientist Intern May 2026 — September 2026
Dialpad Inc. San Ramon, CA

- Developed Agentic AI bots for improved customer service interaction for use in millions of customer service calls
- Worked with GCP BigQuery and built models to mine customer-agent transcripts for agentic skill automation
- Ran quantization, training, and evaluation of Qwen3 models for text chunking, NLU preprocessing using vLLM and LlamaFactory
- Improved LLM performance by 30% with automatic prompt optimization (GEPA) and fine-tuning

Undergraduate Research Assistant January 2024 — October 2025
University of Wisconsin Madison Madison, WI

- Designed large-scale evaluation framework for LLM aggregation across 80k+ prompts and models up to 70B parameters
- Built MARBLES, FUTURE, and ELECPRED datasets to evaluate world-model reasoning beyond pretraining cutoffs
- Ran experiments on HTCondor HPC cluster (8×H200, 8×H100, 26×A100 GPUs) with custom caching and checkpointing
- Demonstrated median-based “Wisdom of Crowds” decoding consistently outperforms greedy and self-consistency
- Co-authored publications (NeurIPS Behavioral ML 2024; EMNLP 2025; ACL/CoLM submissions)

Analytic Software Engineering Intern May 2023 — August 2023
Teradata Inc. San Diego, CA

- Designed and implemented production in-database machine learning feature store used by 100s of enterprise customers
- Authored system design document and architected MVP for backend and frontend independently with modern design patterns
- Built Pandas-like Python SDK and REST API for feature versioning, lineage, and retrieval

Data Science Engineering Intern May 2022 — August 2022
Teradata Inc. San Diego, CA

- Built hyperparameter tuning framework (Grid, Random, Bayesian) integrated into Teradata ML ecosystem
- Implemented Bring-Your-Own-Model capabilities in Java for in-database execution with 10x speedup

SKILLS

Languages	Python, C/C++, Go, Rust, Java, SQL, JavaScript
Tools & Frameworks	PyTorch, XLA, CUDA, JAX, Tensorflow, OpenMP, Deep Learning, React, Node.js, Next.js, NumPy Docker, Kubernetes, Linux, AWS, Git, CI/CD, LLMs, LoRa, Fine-Tuning, Agile, Agentic AI

PROJECTS

ML Compiler Researcher – Learned Operator Fusion December 2025
University of Illinois Urbana-Champaign github.com/sameer-n012/pytorch-learned-fusion

- Replaced heuristic TorchInductor fusion scheduler with learned hardware-aware cost model
- Fine-tuned CodeBERT on 18k torch.fx IR serializations and trained GATv2 graph model on 600 fusion graphs
- Benchmarked 48 Transformer models (up to 15B params); achieved latency gains up to 18ms with compile-time parity

Distributed MapReduce Framework December 2025
University of Illinois Urbana-Champaign

- Implemented Apache Spark-like distributed data processing framework in Go supporting Map, Filter, and Reduce APIs
- Built Cassandra-style distributed file system with 3-node replication and state-based recovery
- Designed centralized scheduler, failure detection and data re-replication via peer-to-peer node recovery

AWARDS & HONORS

Phi Beta Kappa Honor Society	Jan 2024
CheeseHacks 2022 Winner	Nov 2022